Unit absorption asymmetry at prescribed extinction for compact reciprocal dielectrics

A uniform aperture theorem and a sharper joint-resource asymptotic

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Abstract

For any prescribed positive plane-wave extinction cross section E_*, we prove that the supremum of the extinction-normalized absorption difference of opposite time-reversed illuminations is one. This holds for finite compact nonmagnetic reciprocal local anisotropic dielectric bodies, already consisting of two homogeneous layers, at a fixed frequency and with arbitrary fixed pointwise upper bounds $P > 1$ on real permittivity and $Q > 0$ on loss. Every selected body has strictly positive tensor loss, satisfies $X_+ = X_- = E_*$ exactly, and has $\Gamma < 1$; a sequence approaches one. Its size and material resources are unrestricted. The new step beyond the preceding compact-object manuscript is uniform aperture localization after an exact factorization by the squared sine of a physical material-rotation angle. This allows the angle to decrease inversely with aperture while total extinction stays fixed. A second intermediate-value argument restores a full positive loss tensor without changing the target extinction. For the different problem with an enclosing-radius bound, $\|\chi\| \le M$ and $\operatorname{Im}_H \chi \succeq mI > 0$, we also prove a uniform sixth-order remainder and hence $\sup \Gamma = C_*(m, M)(kR)^4 + O_{m,M}((kR)^6)$, where C_* is an explicit jointly realizable electrostatic variational supremum. A reciprocal source Gram bound improves its upper estimate to $C_* \le 4M^3(1 + M)/(9\sqrt{5}\, m^2)$. The numerical value of C_* and the exact finite-resource optimum remain unresolved. No claim of scientific priority or formal proof-assistant verification is made.

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1 Question, scope, and the stronger closure statement

Miller and Monticone's OQ7 asks for fundamental bounds on the redistribution between absorption and scattering under time-reversed illumination [1]. It does not specify a normalization or a resource class. We use free-space plane-wave cross sections, retain all radiated directions and both polarizations, and impose one spatial susceptibility for both experiments. This makes the statement below precise; it does not turn every resource-constrained version of the broad question into the same problem.

Use $e^{-i\omega t}$, vacuum wavenumber $k > 0$, relative permeability I, and relative permittivity $\varepsilon = I + \chi$. The incident electric fields are

$$f_\pm(x, y, z) = e_y e^{\pm ikz}, \qquad Z_0 H_{\mathrm{inc},\pm} = \pm(e_z \times e_y) e^{\pm ikz}. \tag{1}$$

The electric amplitudes have modulus one throughout; their physical intensities are identical. Reciprocity gives the common extinction $X = X_+ = X_- > 0$. With $A_\pm$ and $S_\pm$ denoting absorption and *total* scattering cross sections, define

$$\Gamma = \frac{|A_+ - A_-|}{X} = \frac{|S_+ - S_-|}{X}, \qquad X = A_\pm + S_\pm. \tag{2}$$

All of X, A, S, E_* have units of area. No unit-power Hilbert-space normalization of an infinite plane wave is used.

Theorem 1.1 (Prescribed-extinction supremum)**.** *Fix $k > 0$, $P > 1$, $Q > 0$, and an arbitrary $E_* > 0$. For every $0 < \epsilon < 1$ there is a compact body consisting of two adjacent homogeneous anisotropic dielectric layers with a common finite square aperture, in vacuum, such that*

$$\mu = I, \quad \varepsilon^T = \varepsilon, \quad I \preceq \mathrm{Re}_H\, \varepsilon \preceq PI, \quad 0 \prec \mathrm{Im}_H\, \varepsilon \preceq QI \tag{3}$$

inside both layers, and the pair (1) satisfies

$$X_+ = X_- = E_* \quad \textit{exactly}, \qquad 1 - \epsilon < \Gamma < 1. \tag{4}$$

Consequently $\sup_{X=E_*} \Gamma = 1$ *in this class, and* $\sup_{X=E_*} |A_+ - A_-| = E_*$*. Every selected dimension and coefficient is finite. No common upper bound on aperture, thickness, volume, or integrated material resources, and no common positive lower bound on loss, is imposed.*

The old compact manuscript proved the unrestricted supremum with variable extinction and explicitly left prescribed total extinction unresolved [10]. Simply increasing the aperture of one fixed planar design makes its extinction grow with area. Simply shrinking its physical response afterwards does not follow from that theorem. The present proof establishes the uniform estimate needed for a simultaneous aperture/rotation sequence.

The conclusion is stronger than saying that a small-extinction sequence can have a large ratio. It realizes the same phenomenon at *every* fixed positive E_*. It remains consistent with all fixed-resource bounds: those resources grow along the construction.

Lemma 1.2 (Strict nonattainment)**.** *For every individual compact scatterer in Theorem 1.1, $X > 0$ implies $\Gamma < 1$.*

Proof. Passivity gives $0 \le A_\pm \le X$, so $\Gamma \le 1$. Equality would require one illumination to have absorption X and scattering zero. The far field of a compactly supported L^2 polarization is continuous. A zero integral of its squared modulus therefore makes it zero in every direction. The forward optical theorem, $X = (4\pi/k)\Im(e_y^\dagger F(\pm e_z))$, then gives $X = 0$, a contradiction. □

2 Exact Maxwell formulation with one active material direction

For a bounded domain D, let

$$K_D p = (k^2 I + \nabla\nabla\cdot)(g_k * p)\big|_D, \qquad g_k(x) = \frac{e^{ik|x|}}{4\pi|x|}, \tag{5}$$

with zero extension of p and distributional derivatives. The static contact term is included. The volume equation is $E = f + K_D p$, $p = \chi E$. For any compactly supported polarization,

$$F_p(n) = \frac{k^2}{4\pi}\Pi_n \int_D p(x) e^{-ikn\cdot x}\,\mathrm{d}x, \qquad \Pi_n = I - nn^T, \tag{6}$$

$$\Im\langle p, K_D p\rangle = \frac{k^3}{16\pi^2}\int_{S^2}\left|\Pi_n \int_D p(x) e^{-ikn\cdot x}\,\mathrm{d}x\right|^2 \mathrm{d}\Omega \geq 0. \tag{7}$$

The inner product is conjugate-linear in its first argument. These are exact identities, not a multipole truncation. Boundedness of K_D on $L^2(D)^3$ follows from the static Fourier multiplier $-\xi\xi^T/|\xi|^2$ and the integrable difference-kernel estimate

$$\|K_0\| \leq 1, \qquad \|K_k - K_0\| \leq \frac{3}{4}k^2\left(\frac{3|D|}{4\pi}\right)^{2/3}. \tag{8}$$

For completeness the dynamic difference has radial/tangential eigenvalues $-2b(kr)/(4\pi r^3)$ and $((kr)^2 e^{ikr} + b(kr))/(4\pi r^3)$, where $b(u) = e^{iu}(iu-1)+1$ and $|b(u)| \leq u^2/2$. Schur's test and $\sup_x \int_D |x-y|^{-1}\,\mathrm{d}y \leq 2\pi(3|D|/(4\pi))^{2/3}$ prove (8) without a geometric boundary constant.

Choose thicknesses $h, d > 0$ and square $\mathcal{S}_R = (-R/2, R/2)^2$. The body is $D_R = \mathcal{S}_R \times (0, h+d)$, with absorber $D_{A,R}$ below $z = h$ and plate $D_{W,R}$ above it. Initially use the rank-one susceptibilities

$$\chi_A = i\eta\, e_x e_x^T, \qquad \chi_{W,\theta} = (a + i\delta) u_\theta u_\theta^T, \quad u_\theta = \cos\theta\, e_x + \sin\theta\, e_y, \quad \eta, a, \delta > 0. \tag{9}$$

These are intermediate passive tensors. A final step adds strictly positive loss in their null directions.

Let $J_\theta : L^2(D_R) \to L^2(D_R)^3$ multiply a scalar by e_x in the absorber and by u_θ in the plate. It is a real isometry. Let $\alpha = i\eta$ in the absorber and $\alpha = a + i\delta$ in the plate. Then $\chi = J_\theta \alpha J_\theta^*$ and the exact scalar-current equation is

$$L_{R,\theta} q = J_\theta^* f, \qquad L_{R,\theta} = \alpha^{-1} - J_\theta^* K_{D_R} J_\theta, \qquad p = J_\theta q. \tag{10}$$

The scalar unknown expresses a rank-one *local electric constitutive law*. The electric and magnetic fields reconstructed from it are still three-dimensional Maxwell fields, with all radiation channels retained.

Lemma 2.1 (Uniform inverse and physical solutions)**.** *Put*

$$w = -\Im\alpha^{-1}, \qquad c_0 = \min\left\{\eta^{-1}, \frac{\delta}{a^2+\delta^2}\right\} > 0. \tag{11}$$

For every finite R and every real θ, $L_{R,\theta}$ is invertible and $\|L_{R,\theta}^{-1}\| \leq c_0^{-1}$. Its solution reconstructs the unique outgoing-scattered-field weak Maxwell solution. The inverse bound is independent of aperture and angle.

Proof. By (7),

$$-\Im\langle q, L_{R,\theta} q\rangle = \langle q, wq\rangle + \Im\langle J_\theta q, K_{D_R} J_\theta q\rangle \geq c_0\|q\|^2.$$

The complex Lax–Milgram theorem applied to $iL_{R,\theta}$ gives the inverse, including surjectivity. Reconstruct $E = f + (k^2 I + \nabla\nabla\cdot)(g_k * p)$ and $Z_0 H = (ik)^{-1}\nabla \times E$. The scalar volume potential belongs locally to H^2, so the fields lie in $H_{\mathrm{loc}}(\mathrm{curl})$ and solve Maxwell's equations distributionally. The scattered field is outgoing. Conversely, a weak solution has this volume representation by uniqueness of an entire outgoing vacuum field, and its current lies in the range of J_θ; hence it satisfies (10). The incident field itself is not outgoing. □

2.1 Exact factorization by a physical rotation

For (1) the projected drives satisfy

$$J_\theta^* f_\pm = \sin\theta\, g_\pm, \qquad g_\pm = 1_{(h,h+d)}(z)e^{\pm ikz}. \tag{12}$$

Define the normalized auxiliary responses

$$r_{R,\theta}^\pm = L_{R,\theta}^{-1} g_\pm, \qquad p_{R,\theta}^\pm = \sin\theta\, J_\theta r_{R,\theta}^\pm. \tag{13}$$

The second equality is the actual physical polarization. The first remains well defined at $\theta = 0$, where the physical object is transparent to the incident y polarization. For all angles,

$$X_{R,\theta}^\pm = \sin^2\theta\, Y_R(\theta), \qquad Y_R^\pm(\theta) = k\Im\langle g_\pm, r_{R,\theta}^\pm\rangle, \tag{14}$$

$$A_{R,\theta}^\pm = \sin^2\theta\, B_R^\pm(\theta), \qquad B_R^\pm(\theta) = k\langle r_{R,\theta}^\pm, w r_{R,\theta}^\pm\rangle, \tag{15}$$

$$S_{R,\theta}^\pm = \sin^2\theta\, C_R^\pm(\theta), \qquad C_R^\pm(\theta) = k\Im\langle J_\theta r_{R,\theta}^\pm, K_{D_R} J_\theta r_{R,\theta}^\pm\rangle. \tag{16}$$

Reciprocity of the bilinear pairing gives $Y_R^+ = Y_R^- = Y_R > 0$ even for the auxiliary problem, since $g_- = \overline{g_+}$. The energy identity is $Y_R = B_R^\pm + C_R^\pm$. At $\sin\theta \neq 0$,

$$\Gamma_R(\theta) = \frac{|B_R^+(\theta) - B_R^-(\theta)|}{Y_R(\theta)}. \tag{17}$$

Thus the rotation cancels from the ratio while it controls total extinction. No incident intensity has been changed.

3 A uniform aperture localization theorem

Write $L = h + d$ and $\mathcal{H} = L^2((0, L))$. The symbols J_θ, α, w also denote their one-dimensional versions. Let K_κ be the transverse Fourier fiber of the vector Green operator, and set

$$B_{\kappa,\theta} = J_\theta^* K_\kappa J_\theta, \qquad r_{\infty,\theta}^\pm = (\alpha^{-1} - B_{0,\theta})^{-1} g_\pm. \tag{18}$$

The subscript zero here denotes zero transverse wavevector at the fixed nonzero vacuum frequency; it is not the electrostatic operator. The planar powers $Y_\infty, B_\infty^\pm, C_\infty^\pm$ are given by the same normalized formulas over $(0, L)$, per unit area. Their definitions are auxiliary at $\theta = 0$ and are physical powers divided by $\sin^2\theta$ away from zero.

Theorem 3.1 (Uniform normalized aperture limit)**.** *Fix $k, h, d, \eta, a, \delta > 0$ and a compact interval of angles I_θ. Then, as $R \to \infty$,*

$$\sup_{\theta\in I_\theta} \frac{\|r_{R,\theta}^\pm - r_{\infty,\theta}^\pm\|_{L^2(D_R)}}{R} \longrightarrow 0, \tag{19}$$

where the planar solution is extended constantly in the transverse coordinates. Uniformly on I_θ,

$$\frac{Y_R(\theta)}{R^2} \to Y_\infty(\theta), \qquad \frac{B_R^\pm(\theta)}{R^2} \to B_\infty^\pm(\theta), \qquad \frac{C_R^\pm(\theta)}{R^2} \to C_\infty^\pm(\theta). \tag{20}$$

In particular, if the interval includes zero, these estimates hold for arbitrary sequences $\theta_R \in I_\theta$ with $\theta_R \to 0$. The planar Y_∞ is positive and continuous, so the auxiliary ratios also converge uniformly on that interval.

3.1 Weyl fibers and the grazing circle

With transverse Fourier variable $\kappa \in \mathbb{R}^2$ put

$$\beta(\kappa) = \sqrt{k^2 - |\kappa|^2}, \qquad G_\kappa(z) = \frac{i}{2\beta(\kappa)} e^{i\beta(\kappa)|z|},$$

using the outgoing/decaying branch. For $a, b \in \{1, 2\}$,

$$(K_\kappa)_{ab} = (k^2 \delta_{ab} - \kappa_a \kappa_b) G_\kappa *, \tag{21}$$

$$(K_\kappa)_{a3} = (K_\kappa)_{3a} = i\kappa_a \partial_z G_\kappa *, \tag{22}$$

$$(K_\kappa)_{33} = |\kappa|^2 G_\kappa * - I. \tag{23}$$

The last local term follows from $\partial_z^2 G_\kappa = -\beta^2 G_\kappa - \delta_0$. Schur estimates on the finite interval give

$$\|K_\kappa\| \le C \left(1 + |\kappa| + \frac{k^2 + |\kappa|^2}{|\beta(\kappa)|}\right), \tag{24}$$

$$\|K_\kappa - K_{\mathbf{0}}\| \le C_1 |\kappa|, \qquad |\kappa| \le k/2. \tag{25}$$

The constants depend on k, L only. In particular the grazing singularity has order $||\kappa| - k|^{-1/2}$ and is locally integrable in two dimensions; its square need not be integrable. Since J_θ is an isometry, (24)–(25) hold for the projected fibers uniformly in θ.

At $\kappa = 0$ all active directions are transverse and the kernel is $(ik/2)e^{ik|z-z'|}$ times their dot product. Its imaginary part is positive by the sum of the two outgoing plane-wave fluxes. Thus the same loss coercivity gives $\|r^{\pm}_{\infty,\theta}\| \le c_0^{-1} \|g_\pm\|$ uniformly.

3.2 Proof of Theorem 3.1

Choose any fixed $v \in C_c^\infty((-1/2, 1/2)^2)$ and put $v_R(x) = v(x/R)$. For an arbitrary $r \in \mathcal{H}$ the projected commutator has Fourier form

$$\widehat{H}_{R,\theta}(\kappa) = R^2 \widehat{v}(R\kappa)(B_{\kappa,\theta} - B_{0,\theta}) r. \tag{26}$$

For $|\kappa| \le k/2$, Hilbert-valued Plancherel and (25) bound its inverse transform in global L^2 by $C\|r\|\|\nabla v_R\|_2 = C\|r\|\|\nabla v\|_2$. For $|\kappa| > k/2$ choose $N > 3$. Schwartz decay gives $R^2|\widehat{v}(R\kappa)| \le C_{v,N} R^{2-N} |\kappa|^{-N}$, while

$$\int_{|\kappa| > k/2} |\kappa|^{-N} (\|K_\kappa\| + \|K_{\mathbf{0}}\|) \, \mathrm{d}\kappa < \infty.$$

The inverse Fourier L^1 estimate bounds the high-frequency part in transverse $L^\infty(\mathcal{H})$ by $C_{v,N} R^{2-N} \|r\|$, and on D_R in L^2 by $C_{v,N} R^{3-N} \|r\|$. Therefore

$$\|H_{R,\theta}\|_{L^2(D_R)} \le C_v \|r\| + o(1)\|r\| \tag{27}$$

uniformly in angle. The formula first follows for smooth r; density and boundedness of the finite-domain Green operator extend it to all $r \in \mathcal{H}$. Only the low-frequency piece uses global Plancherel.

For $r = r^{\pm}_{\infty,\theta}$, the trial current $v_R r$ has exact residual $(v_R - 1)g_\pm - H_{R,\theta}$. Apply Lemma 2.1, compare with the constant transverse extension, and use the uniform planar norm bound:

$$\sup_{\theta \in I_\theta} \frac{\|r^{\pm}_{R,\theta} - r^{\pm}_{\infty,\theta}\|}{R} \le C \|1 - v\|_{L^2((-1/2,1/2)^2)} + \frac{C_v}{R} + o(R^{-1}).$$

First let $R \to \infty$ at fixed v, and then approximate one in L^2. This proves (19). Both exact and planar current norms are $O(R)$ uniformly in angle. Cauchy–Schwarz for the work and boundedness of w for its

quadratic form prove the first two limits in (20). The third follows from the exact identities $C = Y - B$ for the finite and planar systems. All-angle scattering is therefore controlled through its exact integrated norm, even if the far field develops a narrowing forward peak. No pointwise angular convergence or discarded edge radiation is assumed.

Finally $B_{0,\theta}$ varies continuously in operator norm with θ; the inverse bound proves continuity of the normalized planar current and powers. Positivity of w and nonzero $g_\pm$ give $Y_\infty(\theta) > 0$. Its minimum on the compact interval is positive, justifying the uniform quotient limit. □

4 A planar design with a regular normalized zero-angle limit

Given $P > 1$, choose an integer $j \geq 1$ such that

$$p_j = \left(\frac{2j}{2j-1}\right)^2 \leq P, \qquad a = p_j - 1 > 0, \qquad kd = (2j-1)\pi. \tag{28}$$

At $\delta = 0$ the plate index along u_θ has phase $2j\pi$ and the orthogonal transverse direction has phase $(2j-1)\pi$. Exact interface matching makes it reflectionless with boundary-plane Jones transmission

$$H_\theta = 2u_\theta u_\theta^T - I_2. \tag{29}$$

The vacuum reference phase of the plate is $e^{ikd} = -1$. The absorber has x permittivity $1 + i\eta$ and is exactly vacuum for y.

Let A_x be the exact scalar slab absorptance for the x polarization. Writing $n = \sqrt{1+i\eta}$, $r_0 = (1-n)/(1+n)$ and $p = e^{inkh}$ gives

$$r_A = \frac{r_0(1-p^2)}{1-r_0^2p^2}, \qquad t_A = \frac{(1-r_0^2)p}{1-r_0^2p^2}, \qquad A_x = 1 - |r_A|^2 - |t_A|^2. \tag{30}$$

For finite $h, \eta > 0$, $0 < A_x < 1$. The flux identity $A_x = k\eta \int_0^h |E_x|^2 \, \mathrm{d}z$ gives strict positivity, and the finite nonzero slab transmission gives strictness below one.

Lemma 4.1 (Exact normalized planar powers)**.** *For the lossless plate, $\delta = 0$, and $\sin\theta \neq 0$,*

$$X_\infty = 4\sin^2\theta, \qquad A_\infty^+ = 0, \qquad A_\infty^- = 4\sin^2\theta\cos^2\theta\, A_x. \tag{31}$$

Consequently the normalized values are $Y_\infty = 4$, $B_\infty^+ = 0$, $B_\infty^- = 4\cos^2\theta A_x$. They extend continuously to $\theta = 0$. For fixed finite h, η, j, the normalized powers and their ratio are jointly continuous in (δ, θ) at $(0,0)$; in particular

$$Y_{\infty,\delta}(0) \longrightarrow 4, \qquad \gamma_{\infty,\delta}(0) := \frac{|B_{\infty,\delta}^+(0) - B_{\infty,\delta}^-(0)|}{Y_{\infty,\delta}(0)} \longrightarrow A_x. \tag{32}$$

Proof. From the absorber side, the y wave crosses the absorber without loss and then the lossless plate. From the plate side, the plate produces an x component of magnitude $|\sin 2\theta|$, whose absorptance is A_x. In either direction the co-polar transmission relative to vacuum is $\cos 2\theta$. The exact planar optical theorem is $X_\infty = 2(1 - \Re t_{\mathrm{co}})$, proving (31).

Continuity cannot be inferred merely by dividing vanishing physical powers. Here is an explicit nonsingular formula. Let r_W, t_W denote the scalar slab coefficients of permittivity $p_j + i\delta$ and thickness d, let $v_W = e^{ikd} = -1$, and put

$$D = 1 - r_A r_W \cos^2\theta, \qquad Y_{\infty,\delta}(\theta) = 2\Re\left\{\frac{1 - t_W/v_W}{D}\right\}. \tag{33}$$

Composition of the two slab scattering matrices gives the co-polar transmission $t_{\mathrm{co}} = 1 + (t_W/v_W - 1)\sin^2\theta/D$. At $\delta = 0$, $r_W = 0, t_W = 1, D = 1$. Every scattered amplitude is $\sin\theta$ times a function continuous in (δ, θ) near $(0, 0)$; explicit expressions are given in Appendix A. Their squared norms give continuous normalized scattering, and $B = Y - C$ gives continuous normalized absorption. This proves (32) with no division by a vanishing quantity. □

Taking $\eta = 1/n$ and $kh = n^2$ makes $A_x \to 1$: the interface reflection tends to zero, while $|p| = \exp[-n^2\Im\sqrt{1 + i/n}] \to 0$. The integer j remains fixed for the prescribed cap P. The absorber thickness increases, but is finite at every chosen accuracy.

5 Fixing total extinction, then making the loss tensor positive

5.1 An exact extinction root in the finite object

First choose a finite absorber with $\eta < Q/3$ and $A_x > 1 - \epsilon/4$. By Lemma 4.1, choose a fixed $0 < \delta < Q/3$ such that $\gamma_{\infty,\delta}(0) > 1 - \epsilon/2$. By continuity there is $\theta_0 > 0$ such that the normalized planar ratio exceeds $1 - 5\epsilon/8$ for $0 \le \theta \le \theta_0$. Now *freeze* these material parameters and thicknesses. Put $Y_0 = Y_{\infty,\delta}(0) > 0$ and set

$$\theta_{\mathrm{lo}}(R) = \sqrt{\frac{E_*}{2R^2Y_0}}, \qquad \theta_{\mathrm{hi}}(R) = \sqrt{\frac{2E_*}{R^2Y_0}}. \tag{34}$$

For sufficiently large finite R this bracket lies in $(0, \theta_0]$. Theorem 3.1 and (14) give

$$X_R(\theta_{\mathrm{lo}}(R)) \to E_*/2, \qquad X_R(\theta_{\mathrm{hi}}(R)) \to 2E_*. \tag{35}$$

The same uniform limit makes $\Gamma_R > 1 - 3\epsilon/4$ throughout the bracket, for sufficiently large R. At each finite R, $L_{R,\theta}$ depends continuously on angle in operator norm, so $X_R(\theta)$ is continuous. The intermediate value theorem supplies at least one angle θ_R in the bracket with $X_R(\theta_R) = E_*$ exactly. No monotonicity or nonzero derivative is required.

Every such root for the rank-one family obeys

$$R\theta_R \longrightarrow \sqrt{E_*/Y_0}, \qquad \Gamma_R(\theta_R) \longrightarrow \gamma_{\infty,\delta}(0). \tag{36}$$

This is the simultaneous limit missing from a fixed-angle aperture theorem.

5.2 Strictly positive tensor loss with the same exact target

For the selected large but finite R, add a common isotropic loss

$$\chi_{A,\nu} = i\eta\, e_x e_x^T + i\nu I, \qquad \chi_{W,\theta,\nu} = (a + i\delta)u_\theta u_\theta^T + i\nu I, \qquad \nu > 0. \tag{37}$$

There is no appeal to an unspecified continuity result at a singular tensor. The unperturbed *full vector electric-field* operator has the exact inverse

$$(I - K_{D_R} J_\theta \alpha J_\theta^*)^{-1} = I + K_{D_R} J_\theta L_{R,\theta}^{-1} J_\theta^*. \tag{38}$$

To check the ordering, use $L_{R,\theta} = \alpha^{-1} - J_\theta^* K_{D_R} J_\theta$ and multiply either side by the operator being inverted. Its norm is at most $1 + \|K_{D_R}\|/c_0$, uniformly over the angle bracket.

The added loss changes this electric-field operator by $-i\nu K_{D_R}$. A Neumann series therefore proves uniform continuity of its inverse as $\nu \downarrow 0$, for this fixed finite R. Current and absorption/extinction quadratic forms are consequently continuous, uniformly over the compact angle bracket. The rank-one common extinction is bounded away from zero on that bracket, so the ratio is also uniformly continuous.

Choose $0 < \nu < Q/3$ sufficiently small that the two strict extinction endpoint inequalities still bracket E_* and that $\Gamma > 1 - \epsilon$ throughout the bracket. A second application of the intermediate value theorem gives a possibly adjusted angle with extinction exactly E_*. The tensor loss eigenvalues are $\{\eta + \nu, \nu, \nu\}$ and $\{\delta + \nu, \nu, \nu\}$, so (3) holds. The real eigenvalues are 1 in the absorber and $\{p_j, 1, 1\}$ in the plate. This proves the existence part of Theorem 1.1; Lemma 1.2 proves nonattainment.

Order of choices. For each accuracy, choose a finite absorber, fix positive active-direction plate loss, establish an angle interval, enlarge the aperture, and only then choose the final isotropic loss and solve the second scalar existence problem. There is no estimate uniform in vanishing loss or diverging thickness. If the scaling in (36) is also desired after isotropic regularization, choose its raw extinction perturbation to be $o(1)$ on the bracket. Mere preservation of the bracket suffices for the main theorem.

5.3 Causality and the resource interpretation

The selected principal-axis values are compatible with passive causal dispersion. A target scalar susceptibility $a + ib$, $a \geq 0, b > 0$, at ω can be realized by

$$\chi(\zeta) = \frac{F}{\omega_0^2 - \zeta^2 - i\gamma\zeta}, \qquad \omega_0^2 - \omega^2 = \frac{a}{b}\gamma\omega, \quad F = \gamma\omega\frac{a^2 + b^2}{b} > 0,$$

for any $\gamma > 0$. Independent oscillators along real principal axes preserve reciprocity and passivity. No bandwidth result follows.

At fixed E_*, passivity makes $1 - \Gamma = (S_h + A_\ell)/E_*$, where h denotes the more absorbing illumination. Thus along a sequence with $\Gamma \to 1$, the more absorbing direction has $A_h \to E_*$ and $S_h \to 0$, while the opposite direction has $A_\ell \to 0$ and $S_\ell \to E_*$. There is no optical-theorem contradiction: its forward radiation peak can narrow as the footprint grows. The co-polar forward-directivity inequality from the preceding paper reads

$$\Gamma \leq 1 - \frac{k^2 E_*}{4\pi \mathcal{D}_{\mathrm{co,h}}}.$$

The construction allows $\mathcal{D}_{\mathrm{co,h}}$ to diverge. It gives no size-independent bound on the resources required for a selected tolerance.

6 Joint resource bounds and an exact asymptotic variational coefficient

The unrestricted construction does not determine the optimum when material resources remain fixed. This section retains the same local susceptibility for both illuminations and sharpens the bounded-loss result in two ways: a source-overlap factor in an exact Maxwell bound, and a remainder uniform over the entire design class. The latter justifies optimizing before taking the small-object limit. The resulting leading coefficient is characterized by an electrostatic variational problem; its numerical optimum is not evaluated.

6.1 Admissible designs and exact operator identities

Fix real unit vectors $e, d \in \mathbb{R}^3$ with $e \cdot d = 0$, and $0 < m \leq M < \infty$. Let $\mathfrak{A}_{m,M}$ consist of positive-volume measurable sets $\Omega \subset B_1(0)$ and measurable tensors $\chi \in L^\infty(\Omega; \mathbb{C}^{3\times 3})$ satisfying, almost everywhere,

$$\chi^T = \chi, \qquad \|\chi\|_{\mathrm{op}} \leq M, \qquad \mathrm{Im}_H\, \chi \succeq mI. \tag{39}$$

The physical body is $R\Omega$, with susceptibility $\chi(x/R)$, vacuum outside, and relative permeability I. Write $t = kR$ and $v = |\Omega|$. Both experiments use this one susceptibility, with incident electric fields

$e\exp(\pm ikd\cdot x)$. Their common extinction is positive. Denote their extinction-normalized absorption difference, equivalently their normalized total-scattering difference, by $\Gamma_t(\Omega,\chi)$.

All operators below act on $\mathcal{H}=L^2(\Omega;\mathbb{C}^3)$, with inner product conjugate-linear in the first argument. Currents are extended by zero before applying the distributional volume potential. Define

$$K_t p=(t^2 I+\nabla\nabla\cdot)(g_t * p)|_\Omega, \qquad g_t(x)=\frac{e^{it|x|}}{4\pi|x|}, \tag{40}$$

$$T_t=(\chi^{-1}-K_t)^{-1}, \qquad W=-\operatorname{Im}_H \chi^{-1}, \tag{41}$$

$$(F_t p)(n)=(I-nn^T)\int_\Omega p(y)e^{-itn\cdot y}\,dy, \qquad Q_t=\frac{F_t^* F_t}{16\pi^2}, \qquad n\in S^2. \tag{42}$$

Thus $\operatorname{Im}_H K_t=t^3 Q_t$ includes every radiation direction. Let $\rho=M^2/m$. The bounded volume-potential and loss estimates give

$$\|K_0\|\le 1, \quad \|K_t-K_0\|\le\frac{3}{4}t^2, \quad W\succeq\rho^{-1}I, \quad \|T_t\|\le\rho, \quad \|Q_t\|\le\frac{v}{6\pi}. \tag{43}$$

These are uniform over measurable supports; the static contact term is part of K_0. In particular, they require no lower bound on volume, component separation, feature size, or boundary regularity. The inverse in (41) is the outgoing weak Maxwell solution operator.

Define the exact work and radiation operators

$$H_t=\operatorname{Im}_H T_t=T_t^*(W+t^3Q_t)T_t, \qquad B_t=T_t^* t^3 Q_t T_t. \tag{44}$$

Reciprocity makes T_t complex symmetric for the bilinear pairing $[u,v]=\int_\Omega u\cdot v$. Consequently H_t is real: it commutes with pointwise complex conjugation. This property is essential below.

Lemma 6.1 (A uniform lower bound for the work operator)**.** *For every $t\ge 0$,*

$$d_t I\preceq H_t\preceq\rho I, \qquad d_t=\frac{m}{[1+M(1+3t^2/4)]^2}. \tag{45}$$

Proof. For an arbitrary input f, put $p=T_t f$ and $E=\chi^{-1}p$. The volume equation gives $f=(I-K_t\chi)E$. Therefore

$$\|f\|\le[1+M(1+3t^2/4)]\|E\|, \qquad \langle f,H_t f\rangle\ge\langle E,(\operatorname{Im}_H\chi)E\rangle\ge m\|E\|^2.$$

This proves the lower bound; the upper bound follows from $\langle f,H_t f\rangle=\operatorname{Im}\langle f,T_t f\rangle\le\rho\|f\|^2$. □

6.2 An exact source-overlap bound

Put $z=d\cdot y$, $c_t=e\cos(tz)$, and $s_t=e\sin(tz)$. Introduce

$$G_t=\begin{pmatrix}\langle c_t,H_t c_t\rangle & \langle c_t,H_t s_t\rangle\\ \langle s_t,H_t c_t\rangle & \langle s_t,H_t s_t\rangle\end{pmatrix}, \qquad \eta_t=\|W^{-1/2}t^3Q_tW^{-1/2}\|, \qquad r_t=\frac{\eta_t}{1+\eta_t}. \tag{46}$$

The matrix G_t is real symmetric and positive semidefinite; its positive trace is the common extinction divided by kR^3.

Theorem 6.2 (Reciprocal Maxwell source overlap)**.** *For every admissible design and $t>0$,*

$$\Gamma_t\le 2r_t\frac{\sqrt{\det G_t}}{\operatorname{tr}G_t}. \tag{47}$$

An upper bound depending only on material resources and a geometric Fourier moment is obtained by defining

$$\mu_t = \left|\frac{1}{v}\int_\Omega e^{2itz}\,dy\right|, \qquad q_t = \frac{1-\mu_t}{1+\mu_t}, \qquad \kappa_t = \frac{\rho}{d_t}, \qquad \overline{\eta}_t = \frac{\rho v t^3}{6\pi}, \tag{48}$$

$$\Psi(x) = \begin{cases} 2\sqrt{x}/(1+x), & 0 \le x \le 1, \\ 1, & x \ge 1. \end{cases} \tag{49}$$

Then

$$\Gamma_t \le \frac{\overline{\eta}_t}{1+\overline{\eta}_t}\Psi(\kappa_t q_t). \tag{50}$$

Proof. The definition of η_t gives $t^3 Q_t \preceq \eta_t W$, hence $0 \preceq B_t \preceq r_t H_t$. The exact two-source power difference is

$$\Gamma_t = \frac{4|\operatorname{Im}\langle c_t, B_t s_t\rangle|}{\operatorname{tr} G_t}. \tag{51}$$

Set $C = H_t^{-1/2} B_t H_t^{-1/2}$. Since $0 \preceq C \preceq r_t I$, its real skew-symmetric imaginary part $J = (C - \overline{C})/(2i)$ has norm at most $r_t/2$. Indeed $-r_t I \preceq C - \overline{C} \preceq r_t I$. For real vectors a, b, subtracting from b its projection on a gives

$$|a^T J b| \le \frac{r_t}{2}\sqrt{\|a\|^2\|b\|^2 - (a\cdot b)^2}.$$

Apply this to $a = H_t^{1/2} c_t$ and $b = H_t^{1/2} s_t$ to prove (47). This is the standard effect-operator distinguishability inequality for a pair of pure states, applied to the Maxwell work inner product; the abstract operator inequality itself is not claimed as new.

For the explicit bound, the ordinary L^2 Gram matrix of c_t, s_t has eigenvalues $v(1+\mu_t)/2$ and $v(1-\mu_t)/2$. Lemma 6.1 bounds the smaller-to-larger eigenvalue ratio of G_t by $\kappa_t q_t$ whenever that is at most one. For a positive 2×2 matrix with eigenvalue ratio $q \in [0,1]$, $2\sqrt{\det G}/\operatorname{tr} G = 2\sqrt{q}/(1+q)$, an increasing function of q. Finally $\eta_t \le \overline{\eta}_t$. Degenerate Gram matrices follow by continuity. □

The common inverse and local constitutive law are retained in (47). Neither this theorem nor its explicit estimate asserts that a current pair saturating the associated operator inequalities is realizable by a local dielectric.

6.3 A sharper upper bound on the quartic coefficient

For each admissible design define

$$q_0 = T_0 e, \qquad q_1 = T_0(ze), \qquad p_j = \int_\Omega q_j(y)\,dy, \tag{52}$$

$$D_0 = \operatorname{Im}(e\cdot p_0) > 0, \qquad \mathcal{C}(\Omega,\chi) = \frac{2|\operatorname{Im}(p_0^* p_1)|}{3\pi D_0}. \tag{53}$$

The affine input ze is an arbitrary L^2 input to the static inverse; it need not be curl-free. Let $\overline{z} = v^{-1}\int_\Omega z$ and $\sigma_z^2 = v^{-1}\int_\Omega (z-\overline{z})^2$.

Theorem 6.3 (Coefficient bound)**.** *Every admissible design satisfies*

$$\mathcal{C}(\Omega,\chi) \le \frac{M^3(1+M)}{3\pi m^2}\, v\sigma_z. \tag{54}$$

Consequently

$$\sup_{\mathfrak{A}_{m,M}} \mathcal{C} \le \frac{4M^3(1+M)}{9\sqrt{5}\,m^2}. \tag{55}$$

Proof. At $t = 0$, $H_0 = T_0^* W T_0$ is real. Moreover,

$$0 \preceq T_0^* Q_0 T_0 \preceq \eta_0^{\mathrm{stat}} H_0, \qquad \eta_0^{\mathrm{stat}} = \|W^{-1/2} Q_0 W^{-1/2}\| \le \frac{\rho v}{6\pi}.$$

Apply the same skew-part argument to real sources e, ze. Writing $D_1 = \langle ze, H_0 ze\rangle$ and $D_{01} = \langle e, H_0 ze\rangle \in \mathbb{R}$ yields

$$\mathcal{C} \le 2\eta_0^{\mathrm{stat}} \sqrt{\frac{D_1 - D_{01}^2/D_0}{D_0}}.$$

The numerator under the square root is the minimum of $\langle ze - ae, H_0(ze - ae)\rangle$ over real a. Taking $a = \overline{z}$ bounds it by $\rho v \sigma_z^2$. The denominator obeys $D_0 \ge d_0 v$, where $d_0 = m/(1+M)^2$. This gives (54). Finally

$$v^2 \sigma_z^2 = v \int_\Omega z^2 - \left(\int_\Omega z\right)^2 \le \frac{4\pi}{3}\frac{4\pi}{15},$$

which proves (55). □

The new upper constant is at most $1/(2\sqrt{5})$ times the earlier $8M^4(1+m)/(9m^3)$ coefficient bound. It is not asserted to be the optimal prefactor.

6.4 Uniform expansion and optimization before taking the limit

Theorem 6.4 (Uniform sixth-order remainder)**.** *Define constants depending only on m, M by*

$$v_* = \frac{4\pi}{3}, \qquad d_0 = \frac{m}{(1+M)^2}, \qquad d_1 = \frac{m}{(1+7M/4)^2}, \tag{56}$$

$$A_D = 2\rho + \frac{3}{4}\rho^2, \qquad A_N = \frac{\rho^3 + (44/45)\rho^2}{\pi}, \qquad K_{m,M} = \frac{v_*}{d_1}\left(A_N + \frac{2\rho^2 A_D}{3\pi d_0}\right). \tag{57}$$

For every $(\Omega, \chi) \in \mathfrak{A}_{m,M}$ and $0 < t \le 1$,

$$\left|\Gamma_t(\Omega,\chi) - \mathcal{C}(\Omega,\chi) t^4\right| \le K_{m,M} t^6. \tag{58}$$

The constant is uniform even when the support and susceptibility depend on t, and even when $|\Omega| \to 0$.

Proof. The resolvent identity and (43) give $\|T_t - T_0\| \le (3/4)\rho^2 t^2$. Put $u = T_t c_t$, $w = T_t s_t$. Taylor's inequalities yield

$$\|u - q_0\| \le U t^2 \sqrt{v}, \qquad U = \frac{\rho}{2} + \frac{3}{4}\rho^2, \tag{59}$$

$$\|w/t - q_1\| \le V t^2 \sqrt{v}, \qquad V = \frac{\rho}{6} + \frac{3}{4}\rho^2. \tag{60}$$

Each of $u, w/t, q_0, q_1$ has norm at most $\rho\sqrt{v}$.

The kernel of Q_t is real and even in t. With $r = x - y$, the identity

$$\int_{S^2} (I - nn^T)(n \cdot r)^2 \, d\Omega_n = \frac{16\pi}{15}|r|^2 I - \frac{8\pi}{15} r r^T$$

and $1 - \cos a \le a^2/2$ imply

$$\|Q_t(x,y) - Q_0(x,y)\|_{\mathrm{op}} \le \frac{t^2 |x-y|^2}{30\pi}.$$

Since $|x-y|\le 2$, Schur's test gives

$$\|Q_t-Q_0\|\le \frac{2v}{15\pi}t^2, \qquad \langle q_0,Q_0q_1\rangle=\frac{p_0^*p_1}{6\pi}. \tag{61}$$

Define

$$N_t=-4\,\mathrm{Im}\langle u,Q_t(w/t)\rangle, \qquad N_0=-4\,\mathrm{Im}\langle q_0,Q_0q_1\rangle.$$

Telescoping the three factors, using (59)–(61), gives

$$|N_t-N_0|\le A_N v^2t^2, \qquad |N_0|\le \frac{2\rho^2}{3\pi}v^2. \tag{62}$$

The numerator constant is explicitly $4[\rho(U+V)/(6\pi)+2\rho^2/(15\pi)]=A_N$.

By bilinear reciprocity the dimensionless common extinction is exactly

$$D_t=\mathrm{Im}\{[c_t,T_tc_t]+[s_t,T_ts_t]\}.$$

The same vector estimates give

$$|D_t-D_0|\le A_D v t^2, \qquad D_t\ge d_1v, \qquad D_0\ge d_0v \quad (0<t\le 1). \tag{63}$$

The exact dimensional scaling of full Maxwell powers is $\Gamma_t=t^4|N_t|/D_t$. Therefore

$$\left|\frac{|N_t|}{D_t}-\frac{|N_0|}{D_0}\right|\le \frac{|N_t-N_0|}{D_t}+\frac{|N_0|\,|D_t-D_0|}{D_tD_0}\le \frac{v}{d_1}\left(A_N+\frac{2\rho^2A_D}{3\pi d_0}\right)t^2.$$

Since $v\le v_*$ and $|N_0|/D_0=\mathcal{C}$, this proves the claim. The displayed volume factors explain why no lower volume fraction is required. The absolute-value inequality also covers $N_0=0$. □

Corollary 6.5 (Exact leading variational coefficient)**.** *Let*

$$\mathcal{F}_{m,M}(t)=\sup_{(\Omega,\chi)\in\mathfrak{A}_{m,M}}\Gamma_t(\Omega,\chi), \qquad \mathcal{C}_*(m,M)=\sup_{(\Omega,\chi)\in\mathfrak{A}_{m,M}}\mathcal{C}(\Omega,\chi). \tag{64}$$

Then

$$\left|\mathcal{F}_{m,M}(t)-\mathcal{C}_*(m,M)t^4\right|\le K_{m,M}t^6 \qquad (0<t\le 1), \tag{65}$$

and, in particular,

$$\lim_{t\downarrow 0}\frac{\mathcal{F}_{m,M}(t)}{t^4}=\mathcal{C}_*(m,M)\le\frac{4M^3(1+M)}{9\sqrt{5}\,m^2}. \tag{66}$$

Proof. The class in (64) is independent of t, and (58) is uniform on that class. Apply $|\sup f-\sup g|\le \sup|f-g|$ and then Theorem 6.3. □

The earlier finite homogeneous unequal-sphere certificate shows that $\mathcal{C}_*(m,M)>0$ for every $0<m\le M$. For every $\varepsilon>0$, one fixed admissible rescaled pattern has $\mathcal{C}>\mathcal{C}_*-\varepsilon$ and hence approaches the leading optimum to that accuracy as $t\downarrow 0$. Conversely, if $t_j\downarrow 0$ and admissible designs satisfy $\mathcal{F}_{m,M}(t_j)-\Gamma_{t_j}(\Omega_j,\chi_j)=o(t_j^4)$, then $\mathcal{C}(\Omega_j,\chi_j)\to\mathcal{C}_*$. These statements do not require a maximizing pattern to exist.

6.5 Equal material cap and loss floor

Corollary 6.6 (A homogeneous-material improvement)**.** *When $M = m$, the susceptibility is necessarily $\chi = imI$ on its support, and*

$$\mathcal{C}(\Omega, imI) \le \frac{m^2}{3\pi} v\sigma_z, \qquad \mathcal{C}_*(m,m) \le \frac{4m^2}{9\sqrt{5}}. \tag{67}$$

Proof. The equality case of $m\|x\|^2 \le \mathrm{Im}(x^*\chi x) \le |x^*\chi x| \le m\|x\|^2$ for all x forces $\chi = imI$. Put $A = -K_0$, so $0 \preceq A \preceq I$ and A is real self-adjoint. Let

$$F = (I + m^2A^2)^{-1}, \qquad U = (I + imA)^{-1}F^{-1/2}.$$

Then U is unitary, with phases in $[-\arctan m, 0]$, $T_0 = imUF^{1/2}$, and $H_0 = mF$. Hence

$$H_0^{-1/2}T_0^*Q_0T_0H_0^{-1/2} = mU^*Q_0U.$$

Because A, Q_0 are real, the norm of its skew imaginary part is $(m/2)\|[Q_0, U^2]\|$. Set $q_{\max} = v/(6\pi)$. The norm of $Q_0 - (q_{\max}/2)I$ is at most $q_{\max}/2$. The spectrum of U^2 lies in a disk of radius $m/\sqrt{1+m^2}$, since its arc has angular width $2\arctan m < \pi$. Centering both operators before estimating the commutator gives

$$\|[Q_0, U^2]\| \le \frac{q_{\max} m}{\sqrt{1+m^2}}.$$

Repeat the real-source Gram argument, now using $m/(1+m^2)\, I \preceq H_0 \preceq mI$. The square root of this condition-number bound cancels $\sqrt{1+m^2}$ above, giving $\mathcal{C} \le 2m^2 q_{\max}\sigma_z$. The same ball moment estimate proves the second inequality. □

6.6 Attainment and constitutive closure remain separate problems

The variational characterization is exact, but none of the explicit upper bounds above is proved physically sharp. In particular, a direct weak-compactness argument does not establish that $\mathcal{C}_*$ is attained. Even without vacuum mixing, the susceptibility cap need not survive a simple laminate effective response. For $M > 1$ and $m < M$, take equal layers of scalar susceptibilities $\chi_\pm = \pm\sqrt{M^2 - m^2} + im$. Continuity of normal displacement and averaging the normal electric field give

$$\chi_{\mathrm{eff},\perp} = im - \frac{M^2 - m^2}{1 + im}, \qquad |\chi_{\mathrm{eff},\perp}|^2 - M^2 = \frac{(M^2-1)(M^2-m^2)}{1+m^2} > 0.$$

Thus an elementary homogenized tensor can lie outside the original cap. This calculation does not prove non-attainment; it identifies a concrete obstruction to assuming constitutive closure. Determining $\mathcal{C}_*$ numerically with certified error, establishing attainment or a precisely described relaxation, and finding the exact finite-t fixed-resource optimum remain unresolved.

Independent algebraic checks. The accompanying script `check_operator_bounds.py` tests the conjugations, factors, source Gram formulas, coefficient bounds, and remainder estimate on 240 reciprocal coercive matrix systems, including volumes spanning eight orders of magnitude. It separately checks 120 homogeneous-material systems, the phase-arc commutator, the spherical fourth moment, and the laminate calculation. All checks passed. These finite-dimensional tests corroborate the operator algebra; they are not a Maxwell discretization or a numerical proof of the continuum statements.

7 Numerical diagnostics and their limits

The proofs above do not rely on a numerical optimizer or discretization. Three distinct numerical checks test the exact planar algebra, finite-aperture implementation, and operator inequalities. Source scripts, raw outputs, and independent audit notes are included in the reproducibility archive.

7.1 Stable planar evaluation and independent layer composition

The expressions in Appendix A evaluate the normalized powers directly, including their auxiliary value at zero angle. A separate four-port cascade solve, performed for 100 randomly selected finite parameters and nonzero angles, agrees in normalized extinction, absorption, and scattering to a maximum absolute discrepancy of 8.04×10^{-14}. Its largest reciprocity norm is 2.50×10^{-16}. An independent implementation checks 20 additional layered cases, including real-permittivity caps of 4 and 1.01; in the lossless-plate cases its maximum discrepancy from the analytic ratio is 1.41×10^{-14}.

Table 1 illustrates near-unit values of the normalized zero-angle planar ratio. These entries use $j = 1$, $\eta = 1/n$, $kh = n^2$, and the small positive plate loss $\delta = 10^{-6}$. Each is a finite-thickness planar calculation. At this fixed positive δ, the large-n ratio has a plate-loss-limited value strictly below one. The unit supremum uses arbitrarily small positive plate loss as well as an increasingly effective absorber, in the order stated in the proof. At physical angle zero the incident y wave induces no current; these are the regular auxiliary limits of powers divided by $\sin^2\theta$, not positive-extinction compact objects. Theorem 3.1 and the two intermediate-value arguments provide the analytic passage to compact bodies at E_*.

Table 1: Exact normalized planar values at the auxiliary angle $\theta = 0$ and $\delta = 10^{-6}$. These are dimensionless per-area quantities.

n	η	kh	$Y_\infty(0)$	$\gamma_\infty(0)$
5	0.20000	25	3.99999805	0.9906314552
10	0.10000	100	3.99999804	0.9993308547
20	0.05000	400	3.99999804	0.9998434035
40	0.02500	1600	3.99999804	0.9999604558
80	0.01250	6400	3.99999804	0.9999897441

7.2 Full-vector finite-aperture discretization at prescribed extinction

For a tractable finite-object diagnostic, set the vacuum wavelength to one, $h = \lambda$, $d = \lambda/2$, $\eta = 0.8$, $a = 3$, and $\delta = 0.02$. Use the intermediate rank-one tensors of (9). A discrete-dipole volume model with cubic cells, the full retarded vector Green interaction, the Clausius–Mossotti self term, and the electric-dipole radiation reaction solves for the scalar active-direction current. The scalar unknown follows from the local rank-one tensor; the propagator and radiation are vectorial. An FFT accelerates finite-domain convolution, with zero padding rather than periodic boundary conditions. The Green-kernel implementation is preserved unchanged from the preceding compact-Maxwell archive.

The rotation angle is tuned until the *discrete* extinction equals $0.1\lambda^2$ within the reported residual. Normalized solves avoid numerical cancellation at small angle. Absorption uses the material-loss quadratic form. Scattering uses the exact all-angle radiation Gram matrix of the discrete electric dipoles, including their self terms. Thus the diagnostic retains radiation outside specular directions. The analytical existence proof uses continuity and the intermediate value theorem, independent of the iteration used for these numerical roots.

Table 2: Finite-grid prescribed-extinction diagnostics. No continuum-error or near-unit optimality certificate is asserted.

R/λ	Cell width	Cells	θ (rad)	X/λ^2	Γ
1	$\lambda/12$	2,592	0.14698137	0.1000000000	0.6072573
2	$\lambda/12$	10,368	0.07650792	0.1000000000	0.7572978
2	$\lambda/16$	24,576	0.07594797	0.1000000000	0.7769269
4	$\lambda/12$	41,472	0.03913265	0.1000000000	0.8442437

Across the completed rows, the largest relative extinction-target error is 1.52×10^{-10}, the largest relative opposite-incidence extinction discrepancy is 5.95×10^{-11}, and the largest normalized linear-system residual is 2.40×10^{-10}. The largest relative optical-theorem closure error is 9.04×10^{-11}. For the one-wavelength aperture, independent far-field integration with 36×72 and 48×96 polar–azimuthal nodes agrees with the radiation Gram quadratic form with maximum relative discrepancy approximately 3.30×10^{-14} over both incidences. These checks establish consistency within the discretization, not its error relative to the continuum.

For the two-wavelength aperture, refining the cell width from $\lambda/12$ to $\lambda/16$ changes Γ from 0.7572978 to 0.7769269, a relative change of 2.59%. This visible discretization dependence precludes treating the tabulated digits as a continuum-converged cross section. For this fixed material and thickness choice the exact continuum planar auxiliary limit is $\gamma_\infty(0) = 0.9492609017$; these particular finite material parameters are not intended to approach one arbitrarily closely. The table also uses zero loss in the inactive tensor directions: the final positive isotropic-loss perturbation is established analytically in Section 5, and is not simulated here.

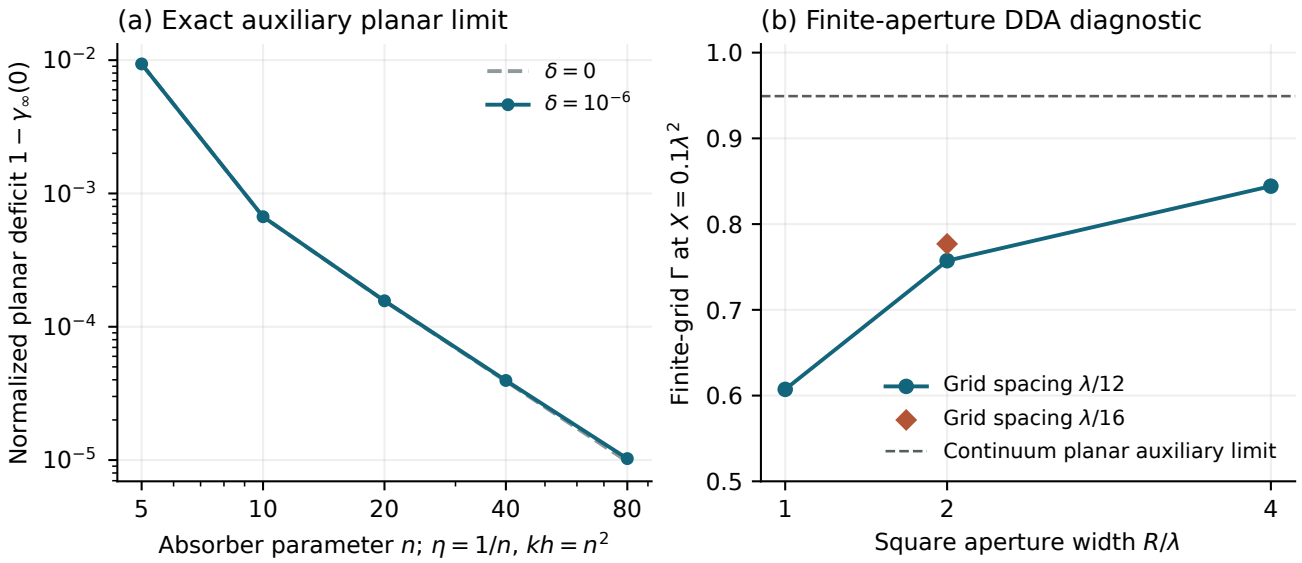


Figure 1: Two separate diagnostics. Left: the lossless-plate auxiliary planar ratio approaches one as the absorber becomes weakly reflecting and thick; fixed positive plate loss leaves a nonzero limiting deficit. Right: finite-grid aperture calculations keep dimensional extinction fixed by adjusting a physical material angle. The isolated refinement point shows visible mesh dependence. The dashed planar line is an analytical reference, not an extrapolation of the finite-grid data.

7.3 Reciprocal operator and coefficient checks

The independent resource-bound script tests 240 reciprocal coercive matrix models for the exact source Gram bound, its geometric estimate, the quartic coefficient inequality, and the stated uniform remainder constants. It also tests 120 homogeneous-corner matrix models for the $M = m$ phase-arc bound, verifies the angular second-moment tensor to maximum error 6.22×10^{-15}, and checks the laminate algebra. All assertions passed. The models retain the reciprocal and positivity identities used by the proof, but they are algebraic checks: no local-Maxwell realization of arbitrary matrix entries or optimization over actual shapes is inferred from them.

8 Prior work, independent audit, and the exact remaining boundary

Extinction symmetry is established by Sounas and Alù [2]; neither that identity nor asymmetric absorption by reciprocal objects is new. Hopkins and coauthors [3] and the direct small-particle expansions discussed in the supplied adversarial audit [13] predate the unequal-sphere mechanism and fixed-smooth-shape quartic upper order. The present manuscript does not rebrand those effects as discoveries.

There are also close layered and metasurface precedents. The reciprocal polarizer component of Kononchuk et al. [8], Sections II–III, has strongly different absorption and reflection from opposite sides; finite-wafer measurements use Gaussian beams. Its polarization extinction ratio is not the dimensional optical-theorem extinction X used here. Máñez-Espina et al. [9], Section 4, derive a maximum directional-absorptance difference in a specified two-port coupled-mode metasurface model without polarization conversion. These results do not state Theorem 1.1. A scalar plate with transmission tending to zero and absorptance difference tending to one has planar extinction density tending to two in the present vacuum reference, and therefore Γ tending to $1/2$. The polarization construction and normalized finite-aperture limit address a different extremal ratio.

Shared-structure constraints also have explicit precedents. Molesky et al. [4], Section II.3, equations (16)–(19), derive off-diagonal source constraints and discuss pointwise enforcement of one geometry. Shim et al. [5] formulate related cross-correlation constraints for multifunctional response. The source Gram inequality used here is a standard positive-operator distinguishability estimate, applied to the exact Maxwell response metric. Its matrix lemma is not claimed new.

The mathematical additions relative to the six supplied manuscripts are: the uniform *normalized* aperture limit, exact positive-extinction tuning and restoration of full tensor loss; and the uniform optimal small-size variational asymptotic with tighter coefficient bounds. The former closes an explicitly outstanding issue of [10]. The latter strengthens [12] but leaves its coefficient optimization genuinely infinite-dimensional.

Two independent AI reviewers rederived the new prescribed-extinction argument, including the zero-angle normalized problem, grazing-circle estimate, inverse identity (38), and both intermediate-value steps. A separate independent check rederived the source Gram and coefficient bounds; the parent independently recalculated the sixth-order constants and volume cancellation. The audit notes and numerical scripts are preserved. These are ordinary mathematical audits, not proof-assistant certificates or independent human peer review. Literature priority is not established by failure to locate an identical theorem.

Precisely specified question	Status of this continuation
Compact local reciprocal dielectric Γ at prescribed $E_* > 0$; fixed pointwise upper caps; unrestricted dimensions	Exact supremum 1, no individual positive-extinction equality; two-layer finite Maxwell construction proved.
Bounded M, positive full tensor floor m, $kR \to 0$, real linear time-reversed polarization	$\sup \Gamma = C_* t^4 + O(t^6)$ uniformly; $0 < C_* < \infty$; an explicit improved upper estimate on C_*.
Numerical value and physical maximizer of $C_*(m, M)$	Unresolved; the variational characterization is not an evaluated optimum.
Exact finite-size resource-constrained optimum, or a common positive loss floor in the $\Gamma \to 1$ construction	Unresolved; not implied by the prescribed-extinction theorem.
All-isotropic compact realization of the prescribed-extinction construction; finite-bandwidth sharpness	Not established here. The construction is locally anisotropic and single-frequency.

A Stable normalized planar amplitudes

Let $s = \sin\theta$, $c = \cos\theta$, $v_A = e^{ikh}$, $v_W = e^{ikd}$, $\Delta = t_W - v_W$, and $D = 1 - r_A r_W c^2$. Order scattered amplitudes as (Lx, Ly, Rx, Ry), subtract the vacuum transmitted wave, and divide by s. The resulting vectors are

$$b_+ = \begin{pmatrix} r_W v_A t_A c/D \\ r_W v_A^2 s/D \\ c v_A(\Delta + v_W r_A r_W)/D \\ s v_A \Delta/D \end{pmatrix}, \tag{68}$$

$$b_- = \begin{pmatrix} c t_A \Delta/D \\ s v_A \Delta/D \\ c\{r_A \Delta(v_W + \Delta c^2)/D + r_W\} \\ s\{r_A \Delta^2 c^2/D + r_W\} \end{pmatrix}. \tag{69}$$

They are continuous at $s = 0$. The normalized scattered powers are $C_\infty^\pm = \|b_\pm\|^2$, and the normalized absorbed powers are $B_\infty^\pm = Y_\infty - \|b_\pm\|^2$, with (33). For the exact lossless plate, $r_W = 0, t_W = 1, v_W = -1$; substitution yields the normalized powers in Lemma 4.1 directly. These formulas avoid floating-point subtraction of a weak response from unit transmission in the numerical validation.

B Independent operator check at zero plate loss

An alternate audit of (32) uses no scattering formula. At $\delta = 0$ the active scalar α is still nonzero in each layer. All active directions are transverse, so $J_\theta^* K_{\mathbf{0}} J_\theta$ has a bounded kernel on a finite interval and is Hilbert–Schmidt. Thus $\alpha^{-1} - J_\theta^* K_{\mathbf{0}} J_\theta$ is Fredholm with index zero. A nullvector reconstructs an outgoing source-free passive one-dimensional Maxwell field. Its flux identity forces both outgoing amplitudes to zero, hence $E = E' = 0$ at an exterior interface. Uniqueness of the finite-layer ordinary differential equation forces the field and current to vanish. The inverse therefore exists, and operator-norm perturbation gives joint continuity in (δ, θ) at $(0, 0)$. Applied to the fixed auxiliary drives $g_\pm$, this independently establishes the regular normalized limit.

C Preservation and reproducibility

The six uploaded PDF manuscripts are preserved byte-for-byte in the source archive, with a SHA-256 manifest. The previously released compact-Maxwell reproducibility archive and quartic adversarial-audit source archive were recovered and retained unchanged. The finite-aperture FFT Green kernel used by the new diagnostic is an unchanged copy from the former, with provenance recorded; the active-direction solver, exact normalized planar formulas, new checks, and this manuscript are separate files. No older proof or claimed formal verification was edited or silently strengthened.

The supplied reproduction runner executes fresh copies of the stable planar and reciprocal-operator checks by default. An optional archive-verification mode checks the preserved audit inputs, and an explicit DDA option runs a selected finite-aperture diagnostic. A separate figure/table generator reads saved JSON evidence without rerunning the numerical solvers. The runner records commands, logs, environment information, validation summaries, and input identities. The full prescribed-extinction result is analytic: mesh truncation and floating-point computations provide diagnostics, not an error certificate for a continuum object arbitrarily close to one.

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